CATASCRIPT
Exploring the catalogs of the British Museum

Final project
Network Analysis course 2025/26

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April 1, 2026 - Paris

Research Question

1. How are romance texts included in vol. 1 from the British Museum distributed across manuscripts?

2. Is it possible to identify any thematic clusters?

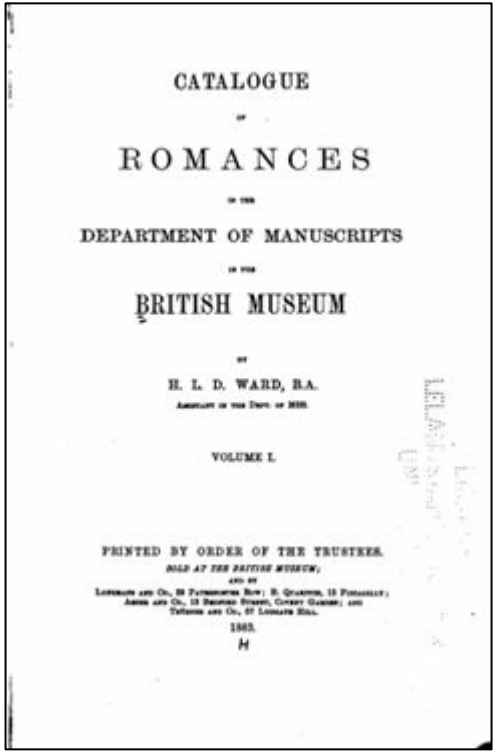
Inspiredy by

1. Martin Grandjean. “A social network analysis of Twitter: Mapping the digital humanities community.” In *Cogent Arts & Humanities*, 2016, 3, pp.1171458. https://doi.org/10.1080/23311983.2016.1171458 .
2. Katarzyna Anna Kapitan, Tarrin Wills. “Sagas and genre: A case for application of network analysis to manuscripts preserving Old Norse-Icelandic saga literature.” In *Digital Scholarship in the Humanities*, 2023, 38, 3, pp. 1130–1144, https://doi.org/10.1093/llc/fqad013

Catalogue manuscript

H. L. D. Ward, BA. *Catalogue of Romances In the Dipartment of Manuscripts in the British Museum*, vol. 1, 1883.

- precise account of the manuscript sources of Romance in the British Museum
- description of works traditionally part of the subject together with “a larger class of literatures which more or less directly has to do with the subject”
- purpose: guide to the Museum Collection and as a handbook to the subject.



CATALOGUE

OF

ROMANCES

IN THE

DEPARTMENT OF MANUSCRIPTS

IN THE

BRITISH MUSEUM

BY

H. L. D. WARD, B.A.

ASSISTANT IN THE DEPT. OF MSS.

VOLUME I.

PRINTED BY ORDER OF THE TRUSTEES.

SOLD AT THE BRITISH MUSEUM;

AND BY

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TRÜBNER AND CO., 57 LUDGATE HILL.

1883.

H

Structure of catalogue

3 sections

- Table of contents
- Index table of manuscripts
- Description of manuscripts

INDEX–TABLE OF MANUSCRIPTS.

mss index | folio index | collection of mss

ROYAL MSS.

		PAGE
4. C. xi. ff. 222–249	Historia Regum Britanniæ	227
——— ff. 280–286 b	Turpin's Chronicle	583
5. F. xv. f. 3 b	Prophecy of Merlin Silvester	313
6. C. viii. ff. 123–133 b	Dares Phrygius	17
6. E. iii. ff. 111 b–112 b	Letters between Alexander and Dindimus	138
7. A. i. ff. 68 b–69 b	———	137
8. B. iv. ff. 19–71 b	Alexandreis	98

TABLE OF CONTENTS.

romance cycle | romance theme

CLASSICAL ROMANCES.

	PAGE
Cycle of Troy :—	
Heroica, by Flavius Philostratus	1
title Iliaca, by Joannes Tzetzes	2
Dictys Cretensis	9
Dares Phrygius	12
Latin Poems, by Simon Chèvre d'Or, etc.	27
Roman de Troie, by Benoît de Sainte-More	35
Historia Trojana, by Guido delle Colonne	40
Trójumanna Saga	62

author

Royal 16. C. iv. A. B.

material | date | size

Notices

Paper; two vols., A.D. 1560 and 1565. Small Quarto; ff. 46, and ff. 98; the full pages of Vol. I. having 35 to 36 and those of Vol. II. 24 to 27 lines of verse.

At the end of the second volume (B) is a copy of the first two and a half chapters of the first book of the Geographia of Agathemerus, in *Greek*, with a *Latin* translation by Pierre Moreau, followed by geographical notes, and verses in *Greek*, *Latin*, and *French*, together with the date of "19 d'Aoust 1579." f. 96 b.

ILIACA: a *Greek* poem, by Joannes Tzetzes, the Byzantine grammarian, who flourished in the middle of the 12th cent.; in

Final dataset

final dataset shape : 622 lines, 228 texts, 394 manuscripts

6 themes of romance: classical romances, british and english traditions, french traditions, miscellaneous romance, appendix, allegorical and didactic romance

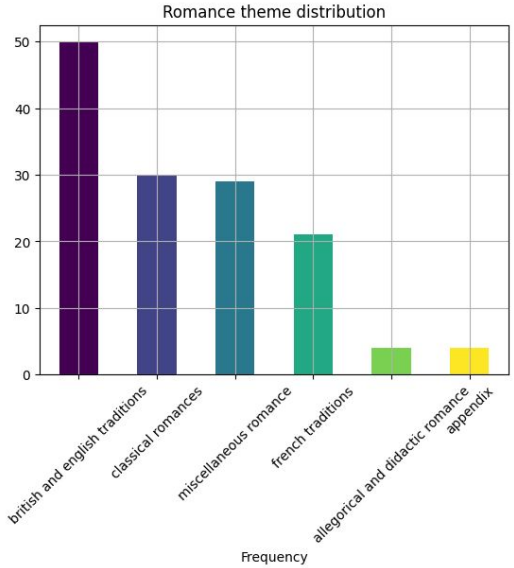
9 collections of manuscripts: royal, cotton, harley, additional, arundel, sloane, lansdowne, burnet, egerton.

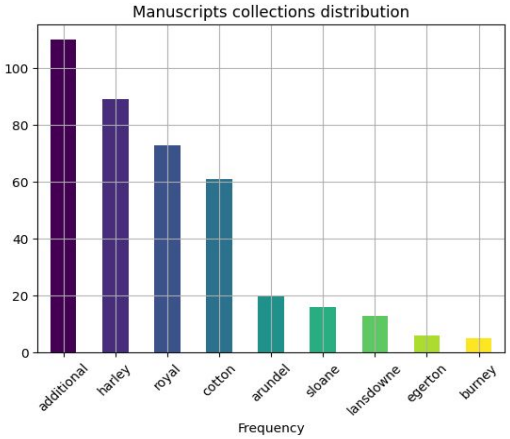
NB. same text can appear several times in different manuscript and they are treated as independent entry;

imbalanced distribution in theme and collections, many missing values due to poor OCR quality.

	title_theme	title_index	author	theme	cycle	collection	mss_ff	mss	ff	page	mss_index	material_class	size_class	cent_i
0	heroica	heroica	flavius philostratus	classical romances	cycle of troy	royal	16. c. xxiii. ff. 2-69 b	16. c. xxiii.	ff. 2-69 b	1	royal 16. c. xxiii. ff. 2-69 b	vellum	quarto	15.0
1	iliaca	iliaca	joannes tzetzes	classical romances	cycle of troy	royal	16. c. iv. a. b.	16. c. iv. a. b.	NaN	2	royal 16. c. iv. a. b.	paper	quarto	16.0
2	iliaca	iliaca	joannes tzetzes	classical romances	cycle of troy	royal	16. d. iii. a. b.	16. d. iii. a. b.	NaN	2	royal 16. d. iii. a. b.	paper	NaN	6.0
3	iliaca	iliaca	joannes tzetzes	classical romances	cycle of troy	harley	harley 5662. ff. 1-56	harley 5662.	ff. 1-56	2	harley 5662. ff. 1-56	paper	quarto	15.0
4	dictys cretensis	dictys cretensis	NaN	classical romances	cycle of troy	harley	harley 3514	harley 3514	NaN	9	harley 3514	vellum	octavo	15.0

Data description





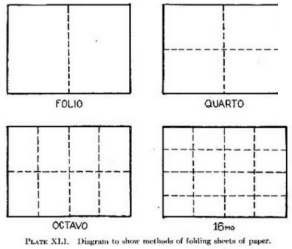
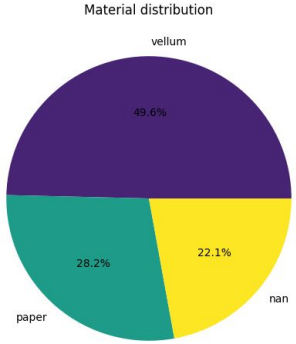
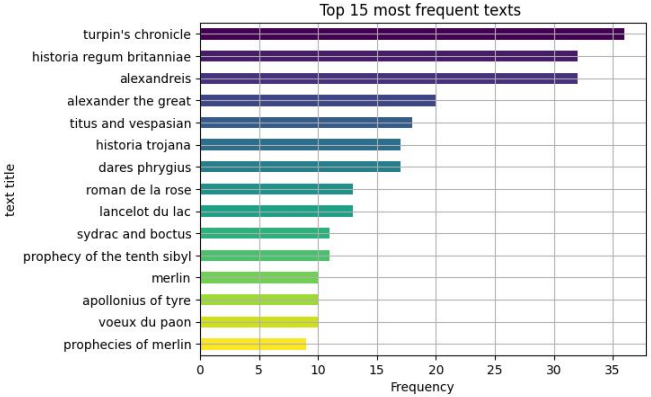
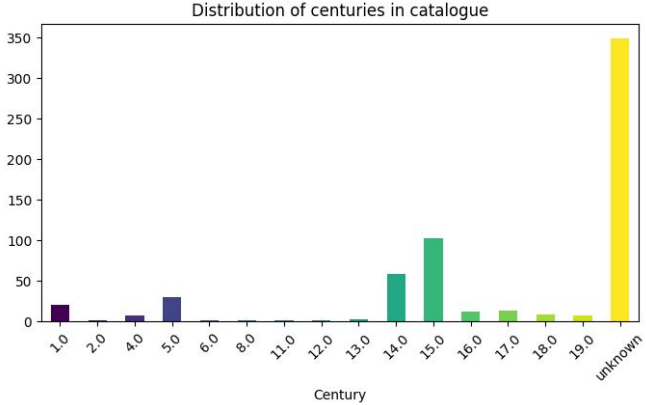
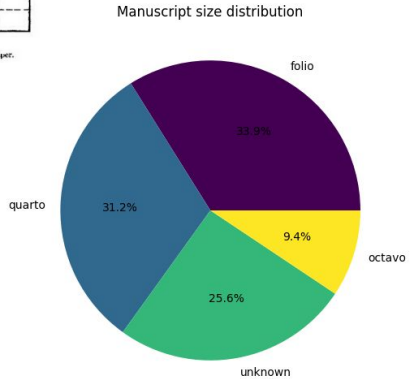


PLATE XLI. Diagram to show methods of folding sheets of paper.









What has been done

Methodology

HOW TO NETWORK

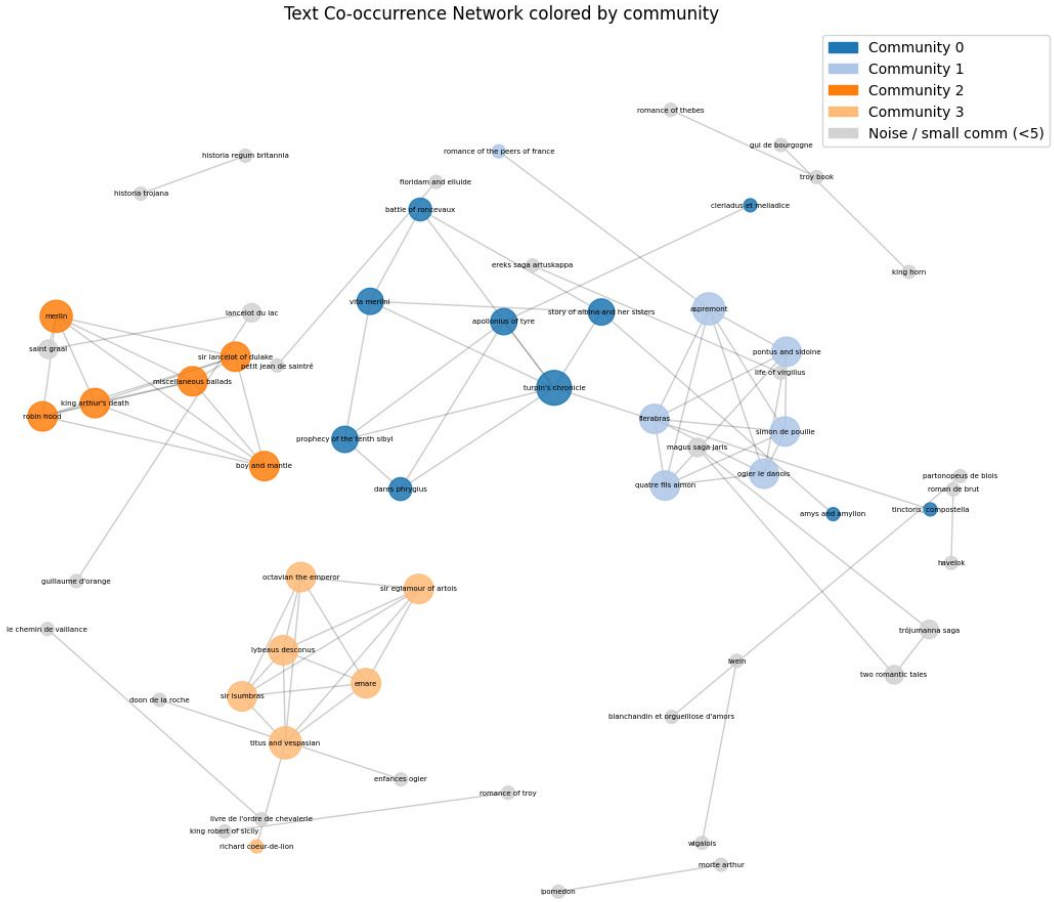
THROUGH MY WORK with the datasets in this project, and with other datasets in other network analysis projects, I have come to a system of processes for taking data from analogue sources to digital networks. The print-to-network data processing pipeline has the following steps:

- **Plan** – Determine the target format or structure.
- **Model** – Identify the structure of the data and decide on a data model.
- **Retrieve** – Obtain machine-readable text.
  For printed text, that means:
  - **Scan** the print pages into a PDF.
  - **OCR** the PDF to create machine-readable text.
- **Segment** – Break the text into meaningful units, and assign an identifier to each.
- **Annotate** – Mark up each segment to capture the relevant metadata, nodes, and edges contained therein.
- **Reshape** – Regularize and reformat the data. Structure it according to the requirements of network analysis tools.
- **Analyze** – Explore the data using statistical methods or visualization.

Network data description

Nodes: texts
Edges: text occurrence within the same manuscript
Cluster: group of texts sharing the same thematic repertoire

Test n. 1: Co-occurrence of texts



Community detection **girvan_newman** and **greedy_modularity_communities** provide similar results.

Pattern identified: texts within the same community tend to appear together

How did this work?

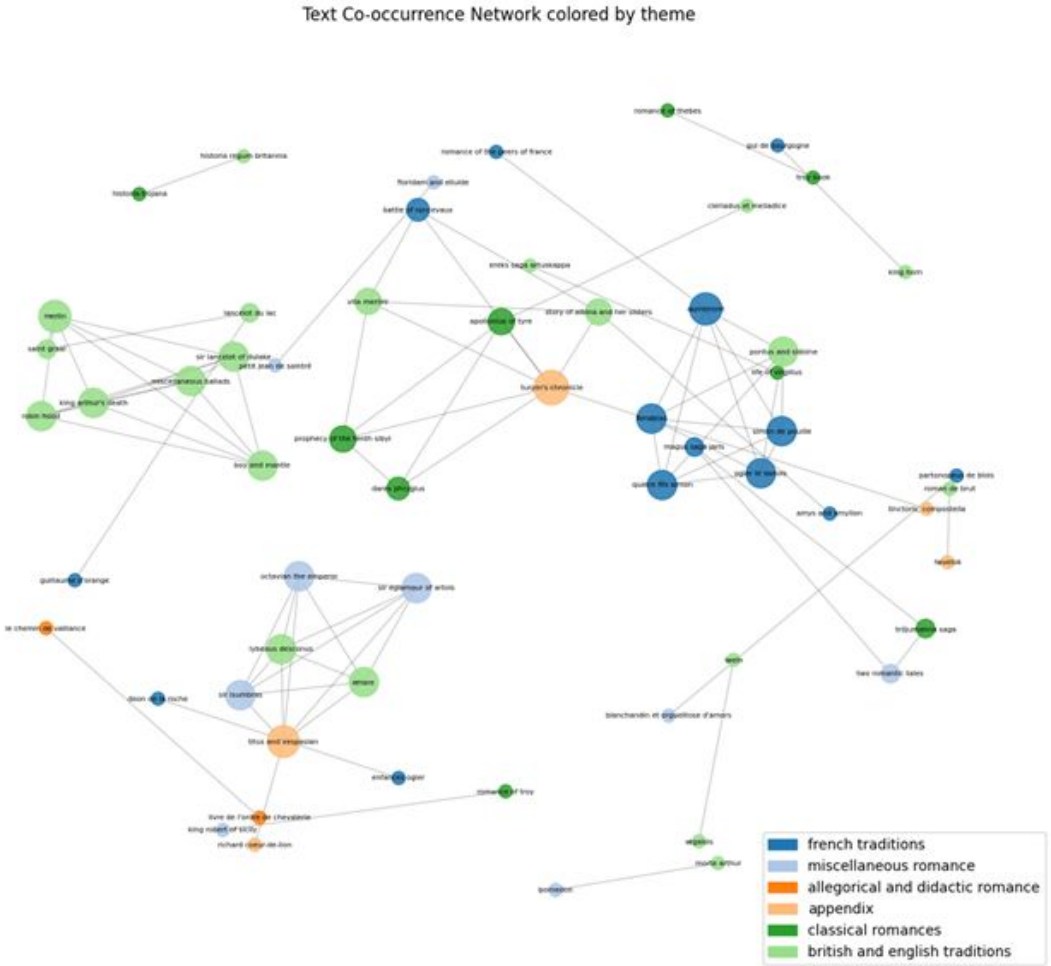
To investigate the underlying factors driving the co-circulation of texts, the nodes were colored according to several attributes.

Among the variables examined (date, size, and material) none produced a visually discernible pattern in the network.

However, **thematic classification** emerged as a meaningful organizing principle: within several communities, **texts sharing the same theme consistently cluster together**.

[*script/test_vis2*]

Interpretation of the results



Texts sharing the same theme consistently cluster together.

Thematic affinity may be a key driver of co-transmission across manuscripts.

What could be done

Enrich the corpus

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KEGAN PAUL, TRENCH, TRÜBNER AND CO., CHARING CROSS ROAD:
AND THE OXFORD UNIVERSITY PRESS, AMEN CORNER, LONDON.

1893.

CATALOGUE

OF

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VOLUME III.

BY

J. A. HERBERT, B.A.,

ASSISTANT IN THE DEPARTMENT OF MSS.

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LONDON.

1910.

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« Previous | **1** of **4** | Next »

Lansdowne MS 348

Reference (shelfmark): Lansdowne MS 348

Title: *Prick of Conscience*

Scope & Content: The manuscript contains:

f. 1r: Notes extracted from Albertus Magnus. Beginning: 'vernena est herba'. Ending: 'et tota domus apperebit plena vestriis'.

f. 2r-127v: *Prick of Conscience* (Main Version IMEV 3428). Beginning: 'Þe myght' of þe fader all' mighti'. Ending: 'Þat for oure loue on Rode can hynge'. Colophon: 'Explicit expliciat. Ludere scriptor eat'.

Missing or misarranged parts of the text (see Lewis and McIntosh, *Descriptive Guide* (1982), pp. 68-69):

- one quire of 8 leaves missing between ff. 57 and 58, containing ll. 3842-4394.

Labs

CATASCRIPT / scripts /

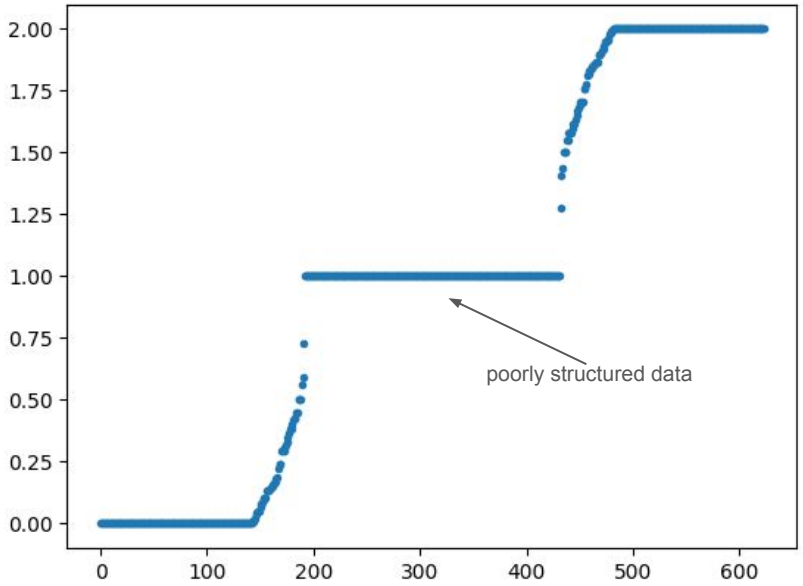
Add file ···

yeliu-dh text coocurrence network 312e19e · 2 hours ago History

Name	Last commit message	Last commit date
..		
__pycache__	Clean dictionary & first visuals	27 days ago
clean_combine_tables	Update test_combine.ipynb	3 weeks ago
tests	Forgot .json (do not use it)	last week
labs.ipynb	text coocurrence network	2 hours ago
manuscripts.py	Clean dictionary & first visuals	27 days ago
test_vis2.ipynb	text coocurrence network	2 hours ago

Implement last labs (5, 6, 7) results. Data exploration.

Beyond visualization.
How data are structured ? *Can we hear the shape of a graph ?*



Scatter plot of nx.laplacian_spectrum(G), normalized; G is bipartite (affiliation network).

NAME	MEANING
coboundary maps	$\delta_k : L^2_\wedge(K_k) \to L^2_\wedge(K_{k+1})$
cochains	elements $f \in L^2_\wedge(K_k)$
cochain complex	$\cdots \longrightarrow L^2_\wedge(K_{k-1}) \xrightarrow{\delta_{k-1}} L^2_\wedge(K_k) \xrightarrow{\delta_k} L^2_\wedge(K_{k+1}) \longrightarrow \cdots$
cocycles	elements of $\ker(\delta_k)$
coboundaries	elements of $\mathrm{im}(\delta_{k-1})$
cohomology classes	elements of $\ker(\delta_k)/\,\mathrm{im}(\delta_{k-1})$
harmonic cochains	elements of $\ker(\Delta_k)$
Betti numbers	$\dim\ker(\Delta_k)$
Hodge Laplacians	$\Delta_k = \delta_{k-1}\delta^*_{k-1} + \delta^*_k\delta_k$
f is closed	$\delta_k f = 0$
f is exact	$f = \delta_{k-1} g$ for some $g \in L^2_\wedge(K_{k-1})$
f is coclosed	$\delta^*_{k-1} f = 0$
f is coexact	$f = \delta^*_k h$ for some $h \in L^2_\wedge(K_{k+1})$
f is harmonic	$\Delta_k f = 0$

Dictionary. Topological jargon from Hodge Laplacian on Graphs

(*) Spectral graph theory (Chung textbook)
(*) Introduction to Analysis on Graphs (Grigor'yan textbook)
(*) Hodge Laplacian on Graphs (https://www.stat.uchicago.edu/~lekheng/work/hodge-graph.pdf)
(*) Community detection in graphs (https://arxiv.org/abs/0906.0612)
(*) Homogeneous Boltzmann-type equations on graphs: A framework for modelling networked social interactions (https://arxiv.org/pdf/2603.25416)
(*) Vertex centrality reconstruction in an inverse problem for information diffusion (https://arxiv.org/pdf/2603.20710)

nx.average_clustering(G) = 0